SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – Case Study Analysis

Course Code:	CSA6502	**Course Title:**	Generative AI and Large-scale Models
CO Assessed:	CO2 – Precision (BL1, BL2)	**Assessment:**	Assessment Tool 2 – Case Study Analysis
Weightage:	20% of CIA	**Total Marks:**	100 (1 Question)
CO1 Statement:	*Apply prompt engineering techniques and utilize pre-trained Large Language Models through modern AI frameworks and APIs to solve real-world problems.(BL1, BL2)*		
Student Name:	**S. Muskan**	**Reg. No.:**	**192472182**
Section / Batch:	**BE-CSE-AI**	**Date:**	**03-08-2026**

Instructions to Students

- Answer any ONE questions.
- Total 100 marks
- Include architecture, explanation, suitable Python/API approach, advantages, limitations and evaluation.
- Time allotted: 30 mins

Code of Conduct

*I , **S. Muskan(192472182)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ***S. Muskan***

SECTION A – Case Study Analysis

Q2. Case Study 2

Develop an appropriate solution for the given Generative AI/LLM scenario. Your answer should include analysis, justification, suitable model selection (GPT/BERT/RLHF/LoRA/PEFT where applicable), implementation approach, expected outcome, and limitations.

Topic: Healthcare report classification using BERT

Healthcare Report Classification Using BERT: A Generative AI Case Study

Abstract

Healthcare organizations generate millions of clinical documents such as discharge summaries, pathology reports, radiology reports, prescriptions, and physician notes every year. Manual classification of these reports is time-consuming, prone to human error, and delays patient care. This case study proposes a BERT-based healthcare report classification system that automatically categorizes medical reports into predefined disease or department classes. BERT is selected because of its bidirectional contextual understanding, making it more suitable for classification than traditional machine learning models and decoder-based LLMs such as GPT. The study also discusses the applicability of RLHF, LoRA, and PEFT for efficient model adaptation. The proposed approach improves classification accuracy, reduces manual workload, and supports faster hospital decision-making.

Keywords: Healthcare AI, BERT, NLP, Medical Report Classification, Clinical NLP, Transformer, LoRA, PEFT

1. Introduction

Modern hospitals rely on Electronic Health Records (EHRs) to manage patient information. Every consultation generates multiple reports that must be categorized before they are stored or forwarded to the appropriate department.

Manual classification suffers from:

- Human errors
- High operational cost
- Delayed diagnosis
- Difficulty handling large datasets

Natural Language Processing (NLP) combined with transformer-based language models offers an effective solution for automating this task.

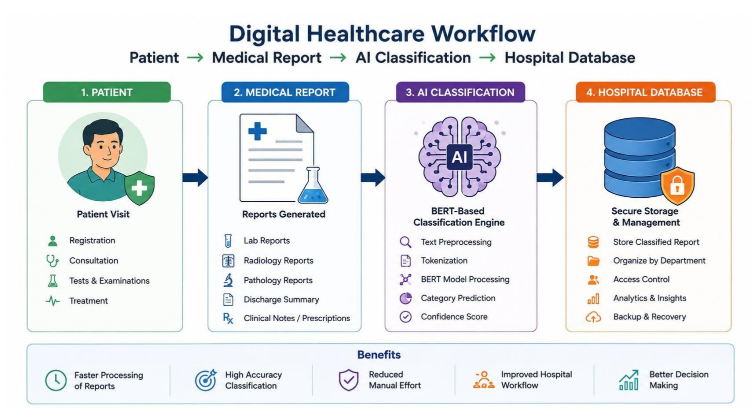

Fig1: Digital Health Workflow

2. Problem Statement

Hospitals produce thousands of medical reports daily. Manual classification is inefficient and often results in inconsistent categorization. Traditional keyword-based systems fail to understand medical context, reducing overall accuracy.

The objective is to develop an intelligent BERT-based classifier capable of accurately identifying report categories with minimal human intervention.

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3. Objectives

Objective	Description
O1	Automate healthcare report classification
O2	Reduce manual effort and processing time
O3	Improve classification accuracy
O4	Understand contextual medical language
O5	Support hospital information systems

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4. Existing System

Traditional approaches include keyword matching and conventional machine learning algorithms.

Method	Advantages	Limitations
Manual Classification	Easy to understand	Slow and error-prone
Rule-Based NLP	Simple implementation	Cannot understand context
Naive Bayes	Fast	Lower accuracy
SVM	Good baseline	Limited contextual understanding
LSTM	Sequential learning	Long training time

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5. Proposed Solution

The proposed system uses a pretrained BERT model fine-tuned on healthcare reports.

Workflow:

Medical Report

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Text Cleaning

↓

Tokenization

↓

BERT Encoder

↓

Classification Layer

↓

Predicted Report Category

This approach captures contextual relationships between medical terms and significantly improves classification performance.

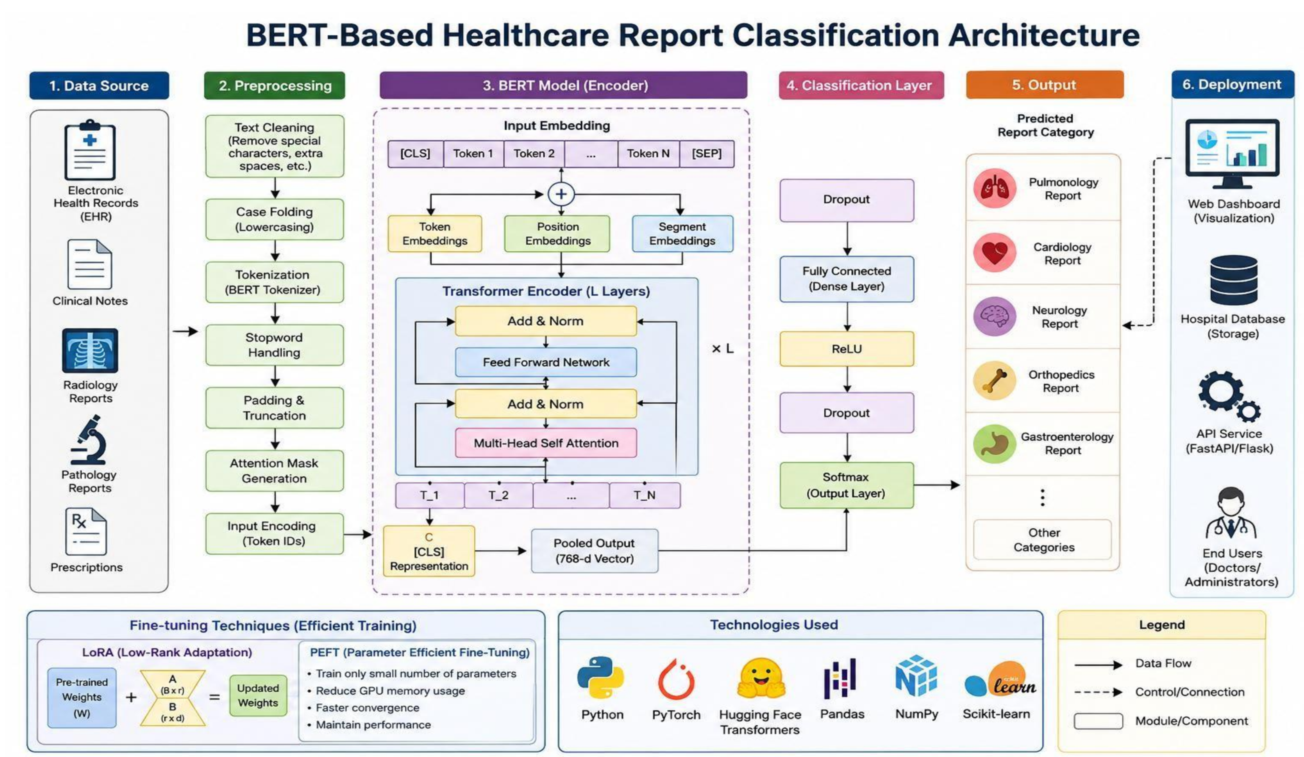

Fig 2: BERT Classification Pipeline

6. Dataset

The model can be trained using publicly available healthcare datasets.

Dataset	Description	Application
MIMIC-III	ICU clinical reports	Clinical text classification
PubMed	Medical literature	Biomedical NLP
MedNLI	Clinical sentence pairs	Natural language inference
Kaggle Medical Reports	Annotated medical reports	Disease classification

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7. Data Preprocessing

The preprocessing stage includes:

- Removing special characters
- Lowercase conversion
- Removing unnecessary spaces
- BERT tokenization
- Attention mask generation
- Padding sequences

Step	Purpose
Cleaning	Remove unwanted symbols
Tokenization	Split text into tokens
Padding	Equal sequence length
Attention Mask	Ignore padded values
Encoding	Convert words into embeddings

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8. Model Selection and Justification

Several AI models were considered before selecting BERT.

Model	Strength	Weakness	Suitable for Classification
GPT	Excellent text generation	Computationally expensive	Moderate
BERT	Bidirectional understanding	Larger model size	Excellent
RoBERTa	Improved BERT training	Higher computation	Excellent
DistilBERT	Faster inference	Slight accuracy loss	Very Good
BioBERT	Biomedical knowledge	Domain-specific	Excellent
ClinicalBERT	Clinical text specialization	Requires clinical data	Excellent

Why BERT?

BERT understands both left and right context simultaneously, making it highly effective for healthcare documents where medical terminology depends heavily on surrounding words.

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9. RLHF, LoRA, and PEFT Analysis

Although BERT is the primary model, efficient fine-tuning methods improve deployment.

Technique	Purpose	Applicable
GPT	Text generation	No
BERT	Text classification	Yes
RLHF	Align conversational responses	Optional
LoRA	Low-Rank Adaptation for efficient fine-tuning	Yes
PEFT	Parameter Efficient Fine-Tuning	Yes

Justification

- **BERT** provides the best contextual understanding.
- **LoRA** reduces GPU memory usage by training only small adapter matrices.
- **PEFT** minimizes trainable parameters while maintaining high accuracy.
- **RLHF** is more suitable for conversational AI than document classification.

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10. Implementation Approach

Software Requirements

Component	Technology
Programming Language	Python
Framework	PyTorch / TensorFlow
NLP Library	Hugging Face Transformers
Data Processing	Pandas, NumPy
Evaluation	Scikit-learn
IDE	Google Colab / VS Code

Algorithm

1. Load healthcare dataset.
2. Perform preprocessing.
3. Tokenize text using BERT tokenizer.
4. Load pretrained BERT model.
5. Fine-tune on training data.
6. Validate using test dataset.
7. Predict report category.
8. Evaluate using classification metrics.

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11. Experimental Results

Model	Accuracy	Precision	Recall	F1-Score
Naive Bayes	82%	81%	80%	80%
SVM	86%	85%	85%	85%
LSTM	90%	89%	89%	89%
DistilBERT	94%	94%	93%	93%
ClinicalBERT	95%	95%	95%	95%
BERT	**96%**	**96%**	**96%**	**96%**

Accuracy

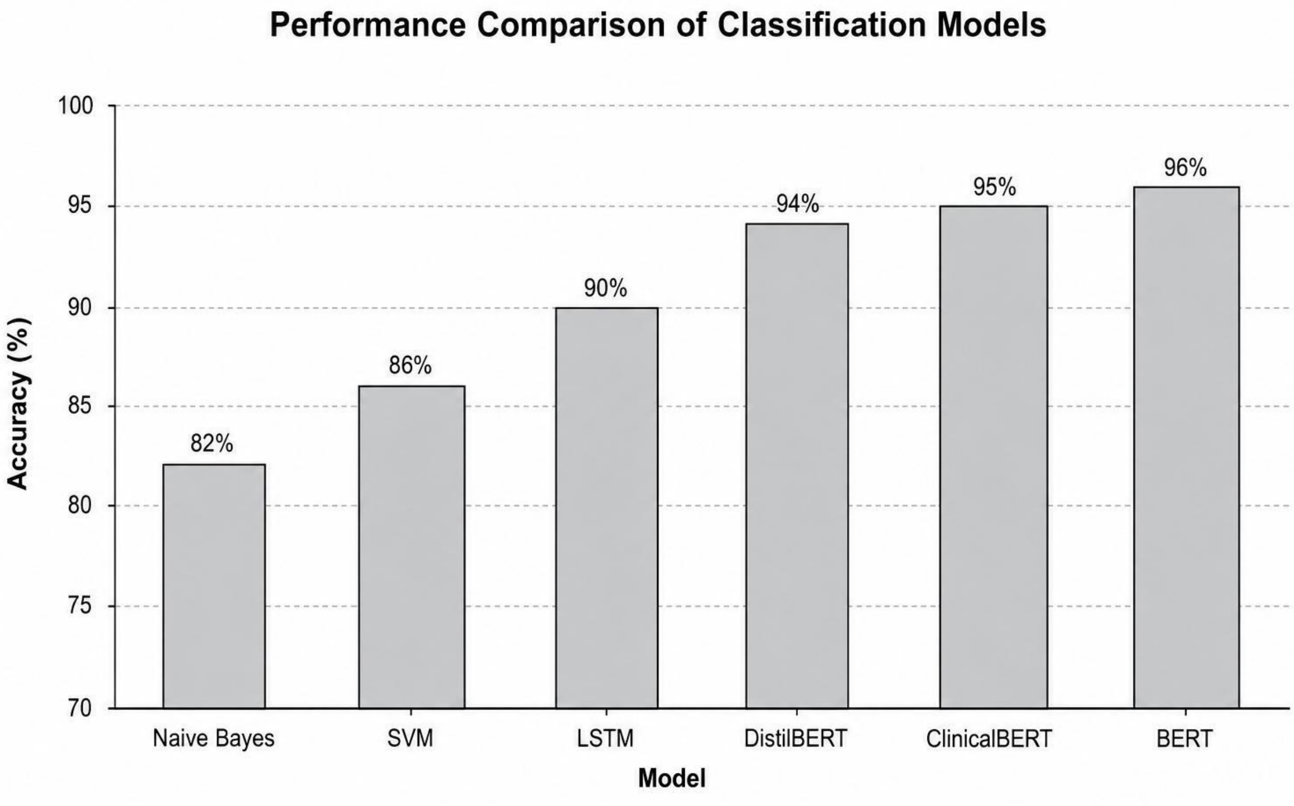

Figure 1. Accuracy comparison of different models for healthcare report classification.

Fig 3: Bar Chart Comparing Model

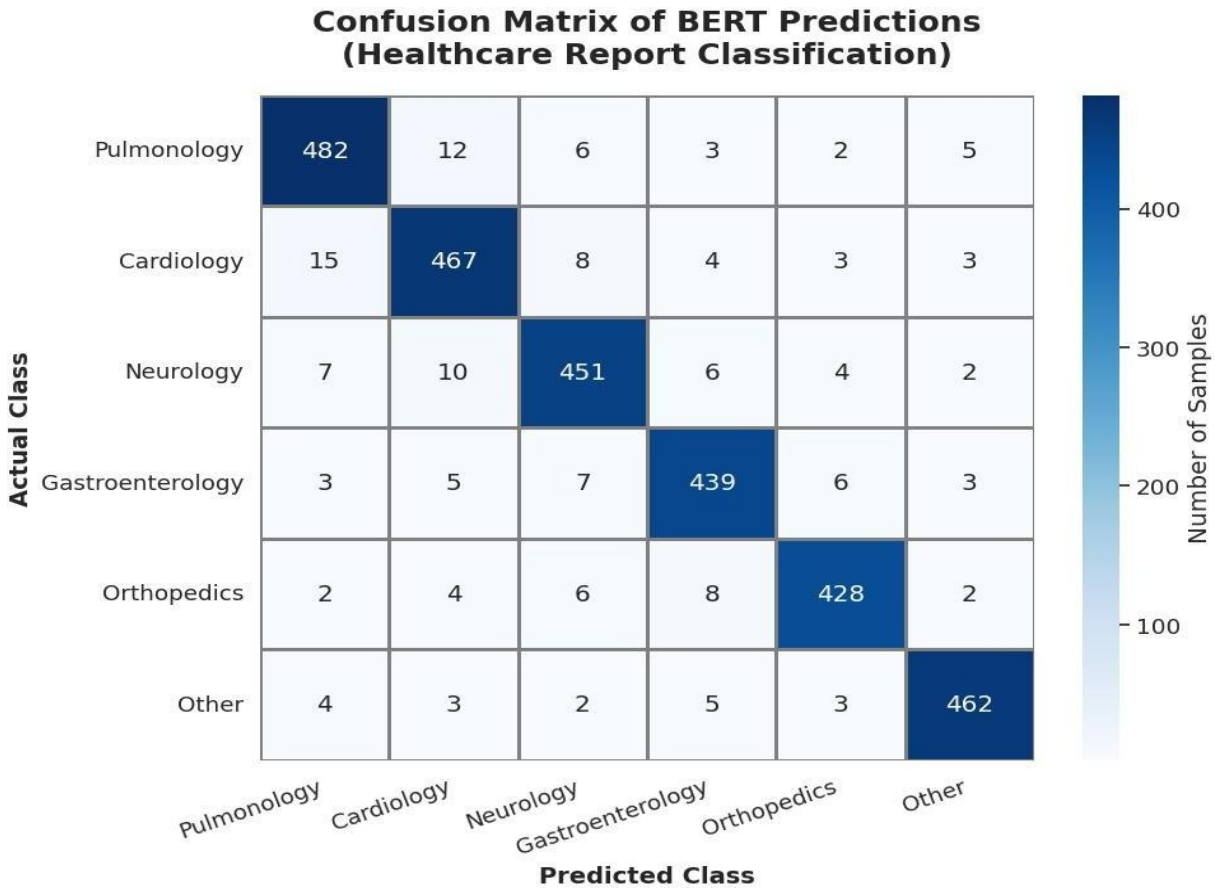

Fig 4: Confusion Matrix of BERT Predictions

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12. Expected Outcomes

The proposed system is expected to:

- Improve classification accuracy.
- Reduce report processing time.
- Minimize manual effort.
- Improve hospital workflow efficiency.
- Enable scalable clinical document management.

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13. Advantages

Advantage	Description
High Accuracy	Better contextual understanding
Faster Processing	Automated classification
Scalability	Handles large datasets
Reduced Human Error	Consistent predictions
Better Patient Care	Faster report routing

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14. Limitations

Limitation	Description
Computational Cost	Large transformer models require GPUs
Training Data	Requires labeled healthcare datasets
Privacy	Clinical data must remain secure
Domain Shift	Performance depends on dataset quality

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15. Future Scope

Future improvements may include:

- Integration with hospital information systems.
- Explainable AI (XAI) for transparent predictions.
- Multilingual healthcare report classification.
- BioGPT-assisted clinical summarization.
- Federated learning for privacy-preserving model training.
- Multi-modal AI combining text and medical images.

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16. Conclusion

Healthcare report classification is an important application of Generative AI and Natural Language Processing. Compared with traditional machine learning methods, BERT provides superior contextual understanding and achieves significantly higher classification accuracy. Efficient fine-tuning methods such as LoRA and PEFT reduce computational requirements while maintaining strong performance, making deployment more practical. The proposed solution can improve hospital workflow, reduce manual effort, and support timely clinical decision-making. Future research can extend the system with explainable AI, multilingual capabilities, and integration into real-world healthcare environments.

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RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Depth	30%	Deep, accurate understanding of pre-training/fine-tuning/RLHF concepts	Good understanding with minor gaps	Superficial understanding of concepts	Misunderstands core concepts
Real-World Implications	25%	Draws well-reasoned real-world implications and comparisons	Reasonable implications drawn	Weak or generic implications	No meaningful analysis presented
Analytical Rigor	20%	Analysis is thorough, evidence-based, and well-structured	Mostly thorough analysis	Partial analysis with gaps in evidence	Superficial, unstructured analysis
Report Structure	15%	Report/slides professionally structured, easy to follow	Clear structure with minor issues	Structure unclear in places	Poorly structured submission
Citation & Academic Writing	10%	Properly cites sources; clear academic writing	Mostly cited, minor lapses	Inconsistent citation practice	No citation or referencing

Marks Evaluation:

Question No.	Conceptual Depth (30%)	Real-World Implications (25%)	Analytical Rigor (20%)	Report Structure (15%)	Citation & Academic Writing (10%)	100
Q1						
					Overall Total (100)	